Homework 4

ORF 245 - Fundamentals of Statistics

Due: Feb. 28th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Joint Discrete RV's) Let the pmf of (X, Y) be given by

	$p(x,y)$	y = 1	2	3	4
	1	$\frac{1}{4}$	0	0	0
x	2	$\frac{1}{8}$	$\frac{1}{8}$	0	0
	3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	0
	4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

(a) Let A denote the event that $X = Y + 1$. Find $\mathbb{P}(A)$.

- **Solution**: First, we list out the possible (x, y) pairs that satisfy the event A: $(2,1), (3,2), (4,3)$. Note that each unique (x, y) pair is a disjoint event. Therefore, by the summation property of disjoint events, we have that

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}((X,Y) = \{(2,1),(3,2),(4,3)\}) \\ &= \mathbb{P}((X,Y) = (2,1)) + \mathbb{P}((X,Y) = (3,2)) + \mathbb{P}((X,Y) = (4,3)) \\ &= \frac{1}{8} + \frac{1}{12} + \frac{1}{16} = \boxed{\frac{13}{48}}\end{aligned}$$

(b) Find the marginal pmf's of X, $p_X(x)$, and Y, $p_Y(y)$.

- **Solution**: To find the marginal PMFs, recall from class that we sum over the rows or columns depending on the variable.

  For X, we sum over the columns to obtain the following table:

x	1	2	3	4
$p_X(x)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

  For Y, we sum over the rows to obtain the following table:

y	1	2	3	4
$p_Y(y)$	$\frac{25}{48}$	$\frac{13}{48}$	$\frac{7}{48}$	$\frac{1}{16}$

(c) Are the two random variables X and Y independent? Justify your answer.

- **Solution**: To show independence, we need to show that for all (x, y) pairs, $\mathbb{P}((X, Y) = (x, y)) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y = y)$. However,

$$\mathbb{P}((X, Y) = (1, 1)) = \frac{1}{4}$$
$$\mathbb{P}(X = 1) \cdot \mathbb{P}(Y = 1) = \frac{1}{4} \cdot \frac{25}{48} \neq \frac{1}{4}.$$

Therefore, X and Y are not independent.

(d) Find conditional pmf of X given $Y = 4$, $p_{X|Y}(x|4)$, and the conditional pmf of X given $Y = 1$, $p_{X|Y}(x|1)$.

- **Solution**: Using the definition of conditional probability, we have that

$$p_{X|Y}(x|4) = \frac{\mathbb{P}((X, Y) = (x, 4))}{\mathbb{P}(Y = 4)} = 16 \cdot \mathbb{P}((X, Y) = (x, 4))$$

From the above formula, we have the following conditional pmf table:

x	1	2	3	4
$p_{X\|Y}(x\|4)$	0	0	0	1

Likewise,

$$p_{X|Y}(x|1) = \frac{\mathbb{P}((X, Y) = (x, 1))}{\mathbb{P}(Y = 1)} = \frac{48}{25} \cdot \mathbb{P}((X, Y) = (x, 1))$$

From the above formula, we have the following conditional pmf table:

x	1	2	3	4
$p_{X\|Y}(x\|1)$	$\frac{12}{25}$	$\frac{6}{25}$	$\frac{4}{25}$	$\frac{3}{25}$

Exercise: 2. (Joint Continuous RV's) Consider the random variables X, Y with the following joint PDF:

$$f(x, y) = \begin{cases} 24xy, & 0 \leq x \leq 1,\ 0 \leq y \leq 1,\ x + y \leq 1 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Note carefully the range of these random variables – both X and Y must be between $[0, 1]$, **and** their sum must be at most one.

(a) Show that the above satisfies the properties of a joint PDF.

- **Solution**: We first verify $f(x,y) \geq 0$. For $0 \leq x \leq 1$, $0 \leq y \leq 1$ and $x+y \leq 1$,

$$f(x,y) = 24xy \geq 0$$

because $x, y \geq 0$. Now, for (x,y) that do not satisfy the constraints, $f(x,y) = 0 \geq 0$. Therefore, $f(x,y) \geq 0$.
Now, we verify that $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y) = 1$. Note the constraint $x+y \leq 1$ is the same as $x \leq 1-y$. Using this fact, we can rewrite the integral as

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy &= \int_0^1\int_0^{1-y} 24xydxdy \\ &= \int_0^1 12(1-y)^2 ydy \\ &= 1\end{aligned}$$

(Please make sure you understand how to set up and integrate double integrals of this form.)
Since there is no assumed information about the marginal PDFs of X and Y, we can define the joint probabilities $\mathbb{P}(a_1 \leq X \leq a_2 \text{ and } b_1 \leq Y \leq b_2) = \int_{a_1}^{a_2}\int_{b_1}^{b_2} f(x,y)dxdy$, so property three is verified by definition.

(b) Find the marginal PDF of X, $f_X(x)$.

- **Solution**: From the definition of a marginal PDF, we have that for $0 \leq x \leq 1$,

$$\begin{aligned}f_X(x) &= \int_{-\infty}^{\infty} f(x,y)dy \\ &= \int_0^{1-x} 24xydy \\ &= 12x(1-x)^2\end{aligned}$$

For x outside this interval, $f_X(x) = 0$.

(c) Find the marginal PDF of Y, $f_Y(y)$.

- **Solution**: Similarly, for $0 \leq y \leq 1$,

$$\begin{aligned}f_Y(y) &= \int_{-\infty}^{\infty} f(x,y)dx \\ &= \int_0^{1-y} 24xydx \\ &= 12y(1-y)^2\end{aligned}$$

For y outside this interval, $f_Y(y) = 0$. Note the symmetry with $f_X(x)$.

(d) Are X and Y independent? Justify your answer.

- **Solution**: Let's examine the effect of the condition $x+y \leq 1$. For (x, y) that are outside this interval, the probability is 0 from the fact that $f(x, y) = 0$ outside that interval. Thus, for any given $X = x$, the choice of Y is constrained to be less than $1 - x$. Now consider the following example, for the intervals $[2/3, 1]$ for both x and y,

$$\mathbb{P}(\{2/3 \leq x \leq 1\} \cap \{2/3 \leq y \leq 1\}) = 0$$

because $x + y > 1$ for every point in this square. However,

$$\mathbb{P}(\{2/3 \leq x \leq 1\}) = \int_{2/3}^{1} 12x(1-x)^2 dx = 1/9$$
$$\mathbb{P}(\{2/3 \leq y \leq 1\}) = \int_{2/3}^{1} 12y(1-y)^2 dy = 1/9$$

Therefore, $\mathbb{P}(\{2/3 \leq x \leq 1\}) \cdot \mathbb{P}(\{2/3 \leq y \leq 1\}) = 1/81 \neq 0$. Since

$$\mathbb{P}(\{2/3 \leq x \leq 1\} \cap \{2/3 \leq y \leq 1\}) \neq \mathbb{P}(\{2/3 \leq x \leq 1\}) \cdot \mathbb{P}(\{2/3 \leq y \leq 1\}),$$

the random variables are not independent. The main point here is that constraints involving two or more variables together will lead to variable dependence.

(e) Find $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

- **Solution**: We use the marginal PDFs and the integral definition of expectation to directly compute these values. For X, we have

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\
&= \int_{0}^{1} x(12x(1-x)^2) dx \\
&= \boxed{\frac{2}{5}}
\end{aligned}$$

For Y, we have

$$\begin{aligned}
\mathbb{E}[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy \\
&= \int_{0}^{1} y(12y(1-y)^2) dy \\
&= \boxed{\frac{2}{5}}
\end{aligned}$$

Remark: Note the symmetry between these two answers. The goal of this entire problem is to recognize that although two variables can have the same distribution, they can still be dependent without one variable being the exact value of the other variable.

Exercise: 3. (Properties of Jointly Distributed Random Variables)
Let X and Y be continuous random variables, with some joint density function $f(x, y)$.

(a) Let $f_Y(y)$ be the marginal pdf of Y. Show that for every value of y satisfying $f_Y(y) > 0$ we have:

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)dx = 1.$$

Be sure to justify each step.

- **Solution**:
  From the definition of the conditional PDF, we have that for $f_Y(y) > 0$,

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} \\ \implies f_{X|Y}(x|y)f_Y(y) &= f(x, y) \end{aligned}$$

  Let's take the integral with respect ot x on both sides, since our original question requires that. Now we have

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dx = \int_{-\infty}^{\infty} f(x, y)dx$$

  From class, the right hand side is $f_Y(y)$ because of the properties of marginal PDFs. Therefore, we can simplify the above equation to

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)f_Y(y)dx = f_Y(y).$$

  Now, note that $f_Y(y)$ is a function of y and does not contain any x terms, so we can pull out that term from the integral. The equation becomes

$$f_Y(y) \int_{-\infty}^{\infty} f_{X|Y}(x|y)dx = f_Y(y)$$

  Since $f_Y(y) > 0$, we can cancel out $f_Y(y)$ from both sides to get

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)dx = 1.$$

(b) Show that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, *without using linearity of expectation.* That is, use only properties of integrals to show the result. Be sure to justify each step.

(Hint: Express the left-hand side as an integral, how can we simplify it or pull terms out?)

- **Solution**: We rewrite $\mathbb{E}[X+Y]$ as a integral:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)dxdy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy$$

Remember that the goal is to get $\mathbb{E}[X]+\mathbb{E}[Y] = \int_{-\infty}^{\infty} xf_X(x)dx + \int_{-\infty}^{\infty} yf_Y(y)dy$. Also, recall that $\int_{-\infty}^{\infty} f(x,y)dx = f_Y(y)$. The second term in our double integral expression looks very close to what we want. Let's pull out the y term from the inner integral because y is not a function of x:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy = \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x,y)dxdy$$

The innermost integral looks very familiar! We can thus simplify:

$$\begin{aligned}\int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f(x,y)dxdy &= \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f(x,y)dx\right) dy \\ &= \int_{-\infty}^{\infty} y \cdot f_Y(y)dy = \mathbb{E}[Y].\end{aligned}$$

Let's try something similar with the first term. The problem is that we have $dxdy$ but we want to have $dydx$. Fortunately, Fubini's theorem says that we can change the order of integration, so now the first double integral becomes

$$\begin{aligned}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dydx \\ &= \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f(x,y)dydx \\ &= \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f(x,y)dy\right) dx \\ &= \int_{-\infty}^{\infty} xf_X(x)dx = \mathbb{E}[X]\end{aligned}$$

Putting it all together, we have

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

as desired.

Exercise: 4. (Properties of Random Sample Mean)
Let$X_1, ..., X_n$ be independent and identically distributed (iid) continuous random variables. That is, their joint probability density function is equal to $f(x_1, ..., x_n) = f_X(x_1) \cdot \cdots \cdot f_X(x_n)$ for some common pdf $f_X(\cdot)$. Note that the marginal pdf of each $X_1, ..., X_n$ is given by the *same* function $f_X(\cdot)$.

Let the mean and variance of each X_i be given by $\mathbb{E}[X_1] = ... = \mathbb{E}[X_n] = \mu_X$, and $\mathrm{Var}(X_1) = ... = \mathrm{Var}(X_n) = \sigma_X^2$, for some values μ_X and $\sigma_X^2 > 0$.

(a) Show that $\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \mu_X$. Be sure to justify each step.

- **Solution**: We use the linearity of expectation.

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^{n}\mu_X = \mu_X$$

We can also use the integral definition of the expectation to directly compute:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty}\frac{1}{n}\sum_{i=1}^{n} x_i f(x_1, x_2, \ldots, x_n)dx_1dx_2\ldots dx_n \\ &= \frac{1}{n}\sum_{i=1}^{n}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} x_i f_X(x_1)f_X(x_2)\cdots f_X(x_n)dx_1dx_2\ldots dx_n\end{aligned}$$

where we used the fact that the variables are independent. Note that $\int_{-\infty}^{\infty} f_X(x_i)dx_i = 1$ by the properties of a PDF, so this sum becomes

$$\frac{1}{n}\sum_{i=1}^{n}\int_{-\infty}^{\infty} x_i f_X(x_i)dx_i = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu_X$$

(b) Show that $\text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{\sigma_X^2}{n}$. Be sure to justify each step.

- **Solution**: We first recall an important fact. From the precept 4, for two independent random variables X and Y, we have that

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y).$$

Now, we return to our original problem. Using the fact that $\text{Var}(aX) = a^2\text{Var}(X)$ for a constant a, we have

$$\text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\text{Var}\left(\sum_{i=1}^{n} X_i\right).$$

Here, from the problem statements, we know that the X_i are independent for all i, so we can use the fact from the precept 4 we mentioned above. Therefore, the expression becomes

$$\begin{aligned} = \frac{1}{n^2}\text{Var}\left(X_1 + X_2 + \ldots + X_n\right) &= \frac{1}{n^2}(\text{Var}(X_1) + \text{Var}(X_2) + \ldots + \text{Var}(X_n)) \\ &= \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}(X_i)\end{aligned}$$

Since each $\text{Var}(X_i) = \sigma^2$, we can substitute in:

$$= \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{1}{n^2}\cdot n\cdot\sigma^2 = \frac{\sigma^2}{n}$$

The important thing to note is that all these calculations are only possible when *all* the X_i are independent. When the independence assumption is removed, it's not possible to have a nice expression for the original variance without more information.

(c) Provide a one-sentence interpretation of the result in part (b) – what happens to the distribution of $\frac{1}{n}\sum_{i=1}^{n} X_i$ as n increases, i.e. as we obtain more and more independent observations coming from the same distribution?

- **Solution**: Since the variance decreases with n, the distribution "concentrates" around the expected value $\mathbb{E}[\frac{1}{n}\sum_{i=1}^{n} X_i]$. In other words, the probability of getting a value that is not close to the expected value decreases when n increases. [Many answers are acceptable here, as long as the answer notes the decrease in variance implies that the distribution will tend towards a constant.]

Exercise: 5. [OPTIONAL] (Joint Continuous RV's II)
Note: This is an ***optional*** exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Suppose random variables X and Y have the following joint PDF:

$$f(x,y) = \begin{cases} 1, & 0 \le x \le 1 \quad \text{and} \quad x \le y \le x+1, \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the marginal distribution of X, $f_X(x)$. What is the name of this family of distributions? What are the associated parameters that characterize the marginal pdf of X?

- **Solution**: We compute the marginal PDF by definition: for $0 \le x \le 1$,

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y)dy \\ &= \int_{x}^{x+1} 1dy \\ &= 1 \end{aligned}$$

For x outside this interval, $f(x,y) = 0$ which implies that $f_X(x) = 0$. This is the *Uniform* distribution, and the associated parameters that characterize the marginal PDF are the endpoints. In this case, the endpoints are 0 and 1, so the marginal PDF is $1/(1-0) = 1$.

(b) Compute $\mu_X = \mathbb{E}[X]$ and $\sigma_X^2 = \mathrm{Var}(X)$.

- **Solution**: Using the integral definition of expectation, we have

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^1 x dx = \boxed{1/2}\end{aligned}$$

Using the decomposition of the variance formula, we have

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \int_{-\infty}^{\infty} x^2 f_X(x) dx - (1/2)^2 \\ &= \int_0^1 x^2 dx - 1/4 \\ &= \boxed{1/12}\end{aligned}$$

(c) Find the marginal distribution of Y, $f_Y(y)$.

(Hint: Break up the marginal distribution into two parts. First find the marginal distribution of Y for $0 \le y \le 1$, and then for $1 < y \le 2$. Think carefully about the limits of integration – a 2-D diagram of the range of X and Y could be useful.)

- **Solution**: Notice that $f(x, y)$ is non-zero only when $0 \le x \le 1$ and $x \le y \le x+1$. This implies that the limits on y such that $f(x, y)$ is non-zero is given by $0 \le y \le 2$. Next, we can express the limits on x in terms of y, and we have two cases. First, if $0 \le y \le 1$, the limits for x are given by $0 \le x \le y$. Second, if $1 \le y \le 2$, then the limits for x are given by $y - 1 \le x \le 1$.
Therefore, when $0 \le y \le 1$,

$$f_Y(y) = \int_0^y 1\, dx = y,$$

and when $1 \le y \le 2$,

$$f_Y(y) = \int_{y-1}^1 1\, dx = (1 - (y - 1)) = 2 - y.$$

In conclusion, the marginal density of Y is given by

$$f_Y(y) = \begin{cases} y, & 0 \le y \le 1, \\ 2 - y, & 1 < y \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(d) What is $\mu_Y = \mathbb{E}[Y]$?

- **Solution**: The expected value is given by:

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y)\, dy.$$

From the marginal density function, we have

$$E[Y] = \int_0^1 y \cdot y\, dy + \int_1^2 y(2-y)\, dy.$$

The first integral is given by

$$\int_0^1 y^2\, dy = \frac{1}{3}y^3\Big|_0^1 = \frac{1}{3}$$

and the second integral is given by

$$\int_1^2 y(2-y)\, dy = \int_1^2 (2y - y^2)\, dy = \frac{2}{3}.$$

Thus,

$$E[Y] = \frac{1}{3} + \frac{2}{3} = 1.$$

(e) Find $\mathrm{Cov}(X, Y)$, the covariance between X and Y.

- **Solution**: We will use the formula $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. From part (b) and (d), we know that $\mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{2} \times 1 = \frac{1}{2}$. Now we will compute $\mathbb{E}[XY]$. Note that

$$E[XY] = \int_0^1 \int_x^{x+1} xy\, dy\, dx.$$

First, we compute the inner integral.

$$\begin{aligned}\int_x^{x+1} xy\, dy &= x \int_x^{x+1} y\, dy \\ &= x \cdot \frac{y^2}{2}\Big|_x^{x+1} = x \cdot \frac{(x+1)^2 - x^2}{2} \\ &= \frac{x(2x+1)}{2}.\end{aligned}$$

Next, we compute the outer integral.

$$\begin{aligned}E[XY] &= \int_0^1 \frac{x(2x+1)}{2}\, dx \\ &= \frac{1}{2}\left(\int_0^1 2x^2\, dx + \int_0^1 x\, dx\right) \\ &= \frac{1}{2}\left(\frac{2}{3} + \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{4+3}{6} = \frac{1}{2} \cdot \frac{7}{6} = \frac{7}{12}\end{aligned}$$

Therefore, $\mathrm{Cov}(X, Y) = \frac{7}{12} - \frac{1}{2} = \frac{1}{12}$.

(f) Are X and Y independent? Justify your answer.

- **Solution**: Note that two continuous random variables are independent if $f(x, y) = f_X(x)f_Y(y)$. Consider $X = Y = 0.5$. Since $f(x, y) = 1$ for $0 \leq x \leq 1$ and $x \leq y \leq x + 1$, we have $f(0.5, 0.5) = 1$. However, by part (a) and (c), we have $f_X(0.5) = 1$ and $f_Y(0.5) = 0.5$, which gives $f_X(0.5)f_Y(0.5) = 0.5 \neq f(0.5, 0.5)$. Therefore, X and Y are not independent.